



3.4 Differentiating Inverse Trigonometric Functions

3.5 Selecting Procedures for Calculating Derivatives

3.6 Calculating Higher Order Derivatives

WARM UP!

Ex 1: Evaluate.

a) $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$	b) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$	c) $\sin^{-1}(-1) = -\frac{\pi}{2}$
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Derivatives of Inverse Trigonometric Functions

$\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$ *See proof below!	$\frac{d}{dx}[\csc^{-1} x] = \frac{-1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx}[\cos^{-1} x] = \frac{-1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\sec^{-1} x] = \frac{1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$	$\frac{d}{dx}[\cot^{-1} x] = \frac{-1}{1+x^2}$



Do you notice a pattern? The derivatives of inverse trigonometric and their corresponding cofunctions form conjugate pairs! Think of these derivatives as BOGO deals.

Ex 2:

If $f(x) = \sin^{-1} x$, then $f'\left(\frac{\sqrt{3}}{2}\right) =$

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{4}{7}$ (D) 2

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f'\left(\frac{\sqrt{3}}{2}\right) = \frac{1}{\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}} = 2$$

Ex 3: Find the derivative.

a) $y = \sin^{-1}(2x)$ $y' = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2 \quad (\text{chain rule})$ $y' = \frac{2}{\sqrt{1-4x^2}}$	b) $f(x) = \sec^{-1}(e^x)$ $f'(x) = \frac{1}{ e^x \sqrt{(e^x)^2-1}} \cdot e^x$ $f'(x) = \frac{e^x}{e^x\sqrt{e^{2x}-1}}$ $f'(x) = \frac{1}{\sqrt{e^{2x}-1}}$ *On the FR portion of the AP exam, answers need not be simplified. The first line would earn full credit!
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Ex 4: Write the equation of the tangent line to $f(x) = \tan^{-1}\left(\frac{x}{2}\right)$ at $x = -2$.

Point

$$f(-2) = \tan^{-1}\left(\frac{-2}{2}\right) = \frac{-\pi}{4}$$

Slope

$$f'(x) = \frac{1}{1 + \left(\frac{x}{2}\right)^2} \cdot \frac{1}{2}$$

$$f'(-2) = \frac{1}{1 + \left(\frac{-2}{2}\right)^2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$y + \frac{\pi}{4} = \frac{1}{4}(x + 2)$$

Ex 5:

If $\lim_{h \rightarrow 0} \frac{\arccos(a+h) - \arccos(a)}{h} = -3$, which of the following could be the value of a ?

$$\frac{-1}{\sqrt{1-a^2}} = -3$$

$$\frac{1}{1-a^2} = 9$$

$$\frac{1}{9} = 1-a^2$$

$$a^2 = \frac{8}{9}$$

$$a = \pm \frac{\sqrt{8}}{3}$$

(A) $\frac{\sqrt{8}}{3}$

(B) $\frac{1}{3}$

(C) 3

(D) 1

**This problem was authored by the Algebras*

Ex 6:

If $\arctan y = \ln x$, then $\frac{dy}{dx} =$

$$\frac{1}{1+y^2} \cdot \frac{dy}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{1+y^2}{x}$$

(A) $\tan\left(\frac{1}{x}\right)$

(B) $\tan(\ln x)$

(C) $\frac{1+y^2}{xy}$

(D) $\frac{x}{1+y^2}$

(E) $\frac{1+y^2}{x}$

**This problem was authored by the Algebras*

Higher Order Derivatives

Higher order derivatives refer to the repeated process of differentiating a function multiple times.

	Lagrange	Leibniz
Function	$f(x)$	y
1 st Derivative	$f'(x)$	$\frac{dy}{dx}$
2 nd Derivative	$f''(x)$	$\frac{d^2y}{dx^2}$
3 rd Derivative	$f'''(x)$	$\frac{d^3y}{dx^3}$
4 th Derivative	$f^{(4)}(x)$	$\frac{d^4y}{dx^4}$
n th Derivative	$f^{(n)}(x)$	$\frac{d^ny}{dx^n}$



Reminder:
 $f^{(4)}(x) \neq f^4(x)$.
 $f^{(4)}(x)$ is the 4th derivative of f
while $f^4(x)$ is the 4th power of f .

Be Careful!
In most cases finding higher order derivatives requires an iterative process. If the n^{th} derivative is incorrect, the $(n+1)^{\text{th}}$ and all other subsequent derivatives will also be incorrect!



Ex 7: Find the indicated derivative.

a) $f(x) = 3x^4 + 2$, $f'''(x) =$

$$f'(x) = 12x^3$$

$$f''(x) = 36x^2$$

$$f'''(x) = 72x$$

b) $y = \sin x$, $\frac{d^2y}{dx^2} =$

$$\frac{dy}{dx} = \cos x$$

$$\frac{d^2y}{dx^2} = -\sin x$$

c) $y = \sin x$, $\frac{d^7y}{dx^7} =$



***Teaching Tip:** Allow the students to figure out the pattern in the derivatives of $\sin x$!

*The n^{th} derivatives of $\sin x$ are cyclical with a 4 term pattern.

$$7 \div 4 = 1 \text{ r } 3 \text{ so } \frac{d^7y}{dx^7} = \frac{d^3y}{dx^3} = -\cos x$$

d) $y = \sin x$, $\frac{d^{2025}y}{dx^{2025}} =$

$$2025 \div 4 = \text{who cares r } 1$$

$$\text{so } \frac{d^7y}{dx^7} = \frac{dy}{dx} = \cos x$$

e) $y = e^x$, $y^{(123456789)} =$

$$y^{(123456789)} = e^x$$

f) $g(x) = \cos x$, $g^{(163)}(x) =$

$$163 \div 4 = \text{who cares r } 3$$

$$g^{(163)}(x) = g'''(x) = \sin x$$

g) $y = \cos(2x)$, $\frac{d^{163}y}{dx^{163}} =$

$$\frac{d^{163}y}{dx^{163}} = 2^{163} \sin(2x)$$

h) $h(t) = \csc t$, $h''(t) =$

$$h'(t) = -\csc t \cot t$$

$$h''(t) = (-\csc t)(-\csc^2 t) + (\csc t \cot t)(\cot t)$$

$$h''(t) = \csc^3 t + \csc t \cot^2 t$$



Ex 8:

If $y = \cos x - \ln(2x)$, then $\frac{d^3y}{dx^3} =$

- (A) $\sin x - \frac{2}{x^3}$
 (B) $-\sin x - \frac{2}{x^3}$
 (C) $\sin x - \frac{1}{x^3}$
 (D) $-\sin x - \frac{1}{x^3}$

$$\begin{aligned}\frac{dy}{dx} &= \sin x - \frac{1}{x} \\ \frac{d^2y}{dx^2} &= \cos x + \frac{1}{x^2} \\ \frac{d^3y}{dx^3} &= -\sin x - \frac{2}{x^3}\end{aligned}$$

Ex 9: Use a graphing utility to find the indicated values.



a) $\frac{d}{dx}(\ln x)|_{x=8}$

NORMAL FLOAT AUTO REAL RADIAN MP
 $\frac{d}{dX}(\ln(X))|_{X=8}$
 0.125000007

$$\frac{d}{dx}(\ln x)|_{x=8} = 0.125$$

b) $f(x) = \arctan x$, $f'(5)$

NORMAL FLOAT AUTO REAL RADIAN MP
 $\frac{d}{dX}(\tan^{-1}(X))|_{X=5}$
 0.0384615399

$$f'(5) = 0.038$$

c) If $y = \frac{3x+2}{\sqrt{x^2+5}}$, $\frac{d^2y}{dx^2}|_{x=\frac{1}{2}}$

NORMAL FLOAT AUTO REAL RADIAN MP
 $\frac{d}{dX}\left(\frac{d}{dX}\left(\frac{3X+2}{\sqrt{XX+5}}\right)\right)|_{X=.5}$
 -0.498783425

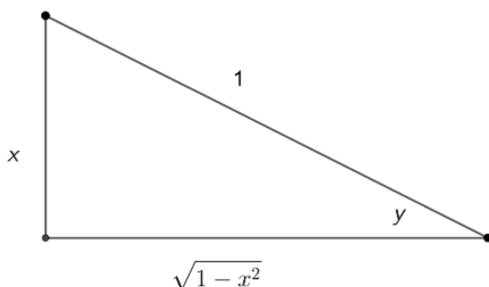
$$\frac{d^2y}{dx^2}\bigg|_{x=\frac{1}{2}} = -0.498 \text{ or } -0.499$$

Strategies for Selecting Procedures for Calculating Derivatives

- Read the problem out loud, reciting all operations.
- Once the operations have been identified, determine what rule or rules are needed to differentiate.
- Decide if its best to rewrite before differentiating.
- Students often forget to use product and or use chain rule incorrectly. Don't be those students! 😊

Proof:

$$\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$$



$$\begin{aligned}y &= \sin^{-1} x \\ \sin y &= \sin(\sin^{-1} x) \\ \sin y &= x \\ \frac{d}{dx}(\sin y) &= \frac{d}{dx}(x) \\ \cos y \frac{dy}{dx} &= 1 \\ \frac{dy}{dx} &= \sec y \\ \frac{dy}{dx} &= \frac{1}{\sqrt{1-x^2}}\end{aligned}$$